



Ravinder Singh

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[Division of Biostatistics](#) | [Department of Population Health Sciences](#) | [Google Scholar](#)

EDUCATION

University Of Utah

Utah, USA

PhD Candidate in Biostatistics, Population Health Sciences

Aug 2021 - Fall 2026 (expected)

Thesis Work: Estimation of Heterogeneous Treatment Effect in Randomized and Non-Randomized Study

Committee: Dr. Tom Greene (primary advisor), Dr. Yizhe Xu, Dr. Jincheng Shen, Dr. Srinivasan Beddhu, Dr. Adam Bress

Skills: Causal Inference, Survival Analysis, Targeted Maximum Likelihood Estimation, Meta Learners, Causal Forest, Propensity Score Weighting/Matching, Risk Modeling, Effect Modeling, Per-Protocol Analyses, Intent-To-Treat Analyses, Doubly-Robust Estimation, Target Trial Emulation, Time-To-Event Analyses, Pseudo-Observations (Jackknife), Cumulative Incidence Functions, Competing Risk Analyses

Bowling Green State University

Ohio, USA

MS in Applied Statistics

Aug 2018 - May 2020

Research Work: Sweet Tweets! Evaluating a New Approach for Probability-Based Sampling of Twitter

[\(Link to article\)](#)

Advisor: Dr. Trent Buskirk

Indian Institute of Science Education and Research

Kerala, India

BS-MS Dual Degree (Mathematics)

Aug 2013 - May 2018

Thesis Work: Basic Study of Copula Theory and its Application in Copula Opinion Pooling Model [\(Link to PPT\)](#)

Advisor: Dr. MP Rajan

RESEARCH PUBLICATIONS

- Sydney E. Hartsell, Guo Wei, **Ravinder Singh**, Michael Throolin, McKenna R. Nevers, Amara Sarwal, Catherine G. Derington, Christa Curtis, Robert E. Boucher, Jincheng Shen, Stavros Drakos, Tom Greene, Srinivasan Beddhu. "Comparative Effectiveness of SGLT2i and GLP-1RA on Kidney and Cardiovascular Outcomes by Kidney Failure Risk." *JASN*. 2026 Jan. [\(Link to article\)](#)
- Catherine G Derington, Amara Sarwal, Guo Wei, Sydney E Hartsell, Michael Throolin, **Ravinder Singh**, McKenna R Nevers, Chong Zhang, Niharika Katkam, Augustine Takyi, Akhil R Chakravartula, Poorvika Babu, Vikrant G Deshmukh, Robert E Boucher, Stavros G Drakos, Tom Greene, Jincheng Shen, Srinivasan Beddhu. "Liraglutide vs Semaglutide vs Dulaglutide in Veterans with Type 2 Diabetes." *JAMA Netw Open*. 2025 Oct. [\(Link to article\)](#)
- Amara Sarwal, **Ravinder Singh**, Guo Wei, Jincheng Shen, McKenna Nevers, Sydney E Hartsell, Catherine G Derington, Augustine Takyi, Akhil R Chakravartula, Niharika Katkam, Robert E Boucher, Stavros G Drakos, Tom Greene, Srinivasan Beddhu. "Glycemic Therapies and the Risk of Gastrointestinal Adverse Events in Veterans with Type 2 Diabetes." *Diabetes Obes Metab*. 2025 Jul 30. [\(Link to article\)](#)
- Lina Brinker, Konstantinos Sideris, Guo Wei, **Ravinder Singh**, McKenna R Nevers, Sydney E Hartsell, Amara Sarwal, Christos P Kyriakopoulos, Eleni Tselioui, Jan Benes, Vojtech Melenovsky, Stavros G Drakos, Thomas C Hanff, Spencer Carter, James C Fang, Monique Cho, Srinivasan Beddhu, Josef Stehlik. "Diabetes Therapy With SGLT2i After Heart Transplant: A Multi-Institutional Analysis." *JACC Heart Fail*. 2025 Jun 25. [\(Link to article\)](#)
- Jincheng Shen, Amara Sarwal, **Ravinder Singh**, Sydney E Hartsell, Guo Wei, McKenna Nevers, Jesse C Christensen, Augustine Takyi, Niharika Katkam, Catherine G Derington, Vikrant G Deshmukh, Robert E Boucher, Stavros G Drakos, Tom Greene, Srinivasan Beddhu. "Comparative Effectiveness of Insulin Glargine, Glucagon-like Peptide-1 Receptor Agonists and Sodium-Glucose Cotransporter-2 Inhibitors in Veterans with Type 2 Diabetes." *Diabetes Obes Metab*. 2025 Apr. [\(Link to article\)](#)

- Amara Sarwal, Robert E. Boucher, Nikita Abraham, **Ravinder Singh**, Xiangyang Ye, Farahnaz Moghaddam, Sydney E. Hartsell, Guo Wei, Srinivasan Beddhu. "Associations of Hyponatremia with Cognition Function and All-Cause Mortality: Post Hoc Analysis of the Systolic BP Intervention Trial." *Kidney360* 4(10):p 1362-1370, October 2023. [\(Link to article\)](#)
- Trent D. Buskirk, Brian P. Blakely, Adam Eck, Richard McGrath, **Ravinder Singh** and Youzhi Yu. "Sweet Tweets! Evaluating a New Approach for Probability-Based Sampling of Twitter." *EPJ Data Sci.* 11, 9 (2022). [\(Link to article\)](#)
- Indravash Chowdhury, **Ravinder Singh**, Christopher Kluse and Mohammad Mayyas. "The Effect of Obstacle Design Architectures on Randomly Ranging AGVs in a Shared Workspace." *Advances in Science, Technology and Engineering Systems Journal*, vol. 6, no. 5, pp. 335-347 (2021). [\(Link to article\)](#)

PUBLICATIONS IN PROGRESS

- Sydney E. Hartsell, Catherine G. Derington, **Ravinder Singh**, Guo Wei, Amara Sarwal, McKenna R. Nevers, Michael Throolin, Ana Mosquera Umbacia, Robert E. Boucher, Jason N. Katz, MD, Terrence Bjordahl, Tahir Zaman, James C. Fang, Alfred K Cheung, Stavros G. Drakos, MD, Tom Greene, Jincheng Shen, Srinivasan Beddhu. "Systolic Blood Pressure and Heart Failure Risk with SGLT2 Inhibitors Versus GLP-1 Receptor Agonists in Type 2 Diabetes." *Under review at Circulation*.
- Shalini Kasera, Sydney E. Hartsell, Guo Wei, Amara Sarwal, McKenna R. Nevers, **Ravinder Singh**, Robert Boucher, Stavros G. Drakos, Catherine G. Derington, PharmD, Srinivasan Beddhu. "Comparative Effectiveness of Synthetic Glucagon-Like Peptide-1 Receptor Agonists and Sodium-Glucose Cotransporter-2 Inhibitors on Weight Loss by Chronic Kidney Disease in Persons with Type 2 Diabetes." *Under review at AJKD*.
- Amara Sarwal, Michael Throolin, Guo Wei, Sydney E. Hartsell, **Ravinder Singh**, Jincheng Shen, McKenna Nevers, Robert Boucher, Tom Greene, Srinivasan Beddhu. "Comparison of common anti-epileptic medications on kidney outcomes and mortality in veterans with type 2 diabetes: valproic acid, lamotrigine, carbamazepine and levetiracetam." *In preparation to CJASN*.

THESIS WORK [PUBLICATIONS IN PROGRESS]

- **Ravinder Singh**, Tom Greene, Srinivasan Beddhu, Yizhe Xu. "R-Learner Based Estimation of Heterogeneous Treatment Effects of SGLT2i vs GLP-1RA on Cardio-Renal Outcomes in Veterans with Type 2 Diabetes." *In preparation*.
- **Ravinder Singh**, Tom Greene, Srinivasan Beddhu, Jincheng Shen. "Targeted-Learning Based Estimation of Heterogeneous Treatment Effects of SGLT2i vs GLP-1RA on Heart Failure Outcomes in Veterans with Type 2 Diabetes." *In preparation*.
- **Ravinder Singh**, Adam Bress, Tom Greene. "Simulation Study to Compare Risk- and Effect- Modeling Approaches for Estimating Individualized Treatment Effects in Randomized Controlled Trials." *In preparation*.

TECHNICAL SKILL SET

R Programming Language | **STATA** | **SQL** | **REDCap** | **SAS** | **Python**

JOB EXPERIENCE

Department of Veteran Affairs
Research Assistant (Without Compensation)

Utah, USA
Jan 2022 - Current

Daiichi Sankyo Inc.
Biostatistics Intern

New Jersey, USA
Jun 2024 - Aug 2024

Bowling Green State University
Assistant Teaching Professor

Ohio, USA
Aug 2019 - Dec 2020

WORKSHOPS AND CONFERENCES

Symposium on Data Science and Statistics (SDSS)

Attended

Utah, USA

Apr 2025

DELPHI Workshop: Machine Learning with R

Teaching Assistant

Utah, USA

Mar 2025

DELPHI Workshop: Advanced R for Data Analysis

Teaching Assistant

Utah, USA

Oct 2024

DELPHI Workshop: Introduction to R for Data Science

Teaching Assistant

Utah, USA

Aug 2024

Joint Statistical Meeting (JSM)

Attended

Oregon, USA

Aug 2024